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Chapter 3

Squares and Square Roots

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Exercise 3(A)

Question 1(i)

The square of -8 is:

1. -16

2. 16

3. -64

4. 64

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Answer

$$\begin{aligned} & (-8)^2 \\ &= (-8) \times (-8) \\ &= 64 \end{aligned}$$

Hence, option 4 is the correct option.**Question 1(ii)** $-\sqrt{18} - \sqrt{11-7}$ is equal to:

1. 4
2. -4
3. 2
4. 0

Answer

$$\begin{aligned} & -\sqrt{18} - \sqrt{11-7} \\ &= -\sqrt{18} - \sqrt{4} \\ &= -\sqrt{18-2} \\ &= -\sqrt{16} \\ &= -4 \end{aligned}$$

Hence, option 2 is the correct option.

Question 1(iii)

Square root of 2.50×10^3 is:

1. 5
2. 125
3. 50
4. none of the above

Answer

$$\begin{aligned}\sqrt{2.50 \times 10^3} \\&= \sqrt{2.50 \times 1000} \\&= \sqrt{2500} \\&= 50\end{aligned}$$

Hence, option 3 is the correct option.

Question 1(iv)

The smallest natural number which on multiplying with 48 gives a perfect square number is:

1. 12
2. 3

3. $\frac{1}{3}$

4. none of the above

Answer

Finding prime factors of 48

$$48 = (2 \times 2) \times (2 \times 2) \times 3$$

Since the prime factor 3 is not in pair.

The given number should be multiplied by 3.

Hence, option 2 is the correct option.**Question 1(v)**

The smallest natural number by which should 175 be divided to get a perfect square number is:

1. 5

2. 7

3. 15

4. none of the above

Answer

Finding prime factors of 175

$$175 = (5 \times 5) \times 7$$

Since the prime factor 7 is not in pair.

The given number should be divided by 7.

Hence, option 2 is the correct option.

Question 2(i)

Find the square of 59.

Answer

$$59^2$$

$$= 59 \times 59$$

$$= 3481$$

$$59^2 = 3481$$

Question 2(ii)

Find the square of 6.3.

Answer

$$6.3^2$$

$$= 6.3 \times 6.3$$

$$= 39.69$$

$$6.3^2 = 39.69$$

Question 2(iii)

Find the square of $15\frac{2}{3}$

Answer

$$\begin{aligned} & \left[15\frac{2}{3}\right]^2 \\ &= \left[\frac{47}{3}\right]^2 \\ &= \frac{47 \times 47}{3 \times 3} \\ &= \frac{2209}{9} \\ &= 245\frac{4}{9} \\ 15\frac{2}{3} &= 245\frac{4}{9} \end{aligned}$$

Question 3(i)

By splitting into prime factors, find the square root of 11025.

Answer

$$\begin{aligned} 11025 &= (3 \times 3) \times (5 \times 5) \times (7 \times 7) \\ \Rightarrow \sqrt{11025} &= 3 \times 5 \times 7 \\ \Rightarrow \sqrt{11025} &= 105 \end{aligned}$$

Square root of 11025 is 105.

Question 3(ii)

By splitting into prime factors, find the square root of 396900.

Answer

$$396900 = (2 \times 2) \times (3 \times 3) \times (3 \times 3) \times (5 \times 5) \times (7 \times 7)$$

$$\Rightarrow \sqrt{396900} = 2 \times 3 \times 3 \times 5 \times 7$$

$$\Rightarrow \sqrt{396900} = 630$$

Square root of 396900 is 630.

Question 3(iii)

By splitting into prime factors, find the square root of 194481.

Answer

$$194481 = (3 \times 3) \times (3 \times 3) \times (7 \times 7) \times (7 \times 7)$$

$$\Rightarrow \sqrt{194481} = 3 \times 3 \times 7 \times 7$$

$$\Rightarrow \sqrt{194481} = 441$$

Square root of 194481 is 441.

Question 4(i)

Find the smallest number by which 2592 be multiplied so that the product is a perfect square.

Answer

Finding prime factors of 2592

$$2592 = (2 \times 2) \times (2 \times 2) \times 2 \times (3 \times 3)$$

Since the prime factor 2 is not in pair.

2 should be multiplied with 2592 so that the product is a perfect square.

Question 4(ii)

Find the smallest number by which 12748 be multiplied so that the product is a perfect square.

Answer

Finding prime factors of 12748

$$12748 = (2 \times 2) \times 3187$$

Since the prime factor 3187 is not in pair.

3187 should be multiplied with 12748 so that the product is a perfect square.

Question 5

Find the smallest number by which 10368 be divided so that the result is a perfect square. Also, find the square root of the resulting number.

Answer

Finding prime factors of 10368

$$10368 = (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times 2 \times (3 \times 3) \times (3 \times 3)$$

Since the prime factor 2 is not in pair.

$$10368 \div 2 = 5184$$

$$\sqrt{5184} = 2 \times 2 \times 2 \times 3 \times 3$$

$$\sqrt{5184} = 72$$

2 should be divided by 10368 so that the 5184 is a perfect square and square root of 5184 is 72.

Question 6(i)

Find the square root of 0.1764.

Answer

$$0.1764 = \frac{1764}{10000}$$

$$\sqrt{\frac{1764}{10000}} = \frac{\sqrt{1764}}{\sqrt{10000}}$$

$$= \frac{\sqrt{(2 \times 2) \times (3 \times 3) \times (7 \times 7)}}{\sqrt{(2 \times 2) \times (2 \times 2) \times (5 \times 5) \times (5 \times 5)}}$$

$$= \frac{2 \times 3 \times 7}{2 \times 2 \times 5 \times 5}$$

$$= \frac{42}{100}$$

$$= 0.42$$

$$\sqrt{0.1764} = 0.42$$

Question 6(ii)

Find the square root of $96\frac{1}{25}$.

Answer

$$\begin{aligned}\sqrt{96\frac{1}{25}} &= \sqrt{\frac{2401}{25}} \\&= \frac{\sqrt{2401}}{\sqrt{25}} \\&= \frac{\sqrt{(7 \times 7) \times (7 \times 7)}}{\sqrt{(5 \times 5)}} \\&= \frac{7 \times 7}{5} \\&= \frac{49}{5} \\&= 9\frac{4}{5} \\\sqrt{96\frac{1}{25}} &= 9\frac{4}{5}\end{aligned}$$

Question 6(iii)

Find the square root of 0.0169.

Answer

$$\sqrt{0.0169} = \sqrt{\frac{169}{10000}}$$

$$= \frac{\sqrt{169}}{\sqrt{10000}}$$

$$= \frac{\sqrt{13 \times 13}}{\sqrt{10 \times 10}}$$

$$= \frac{13}{100}$$

$$= 0.13$$

$$\sqrt{0.0169} = 0.13$$

Question 7(i)

Evaluate:

$$\sqrt{\frac{14.4}{22.5}}$$

Answer

$$\sqrt{\frac{14.4}{22.5}}$$

$$= \frac{\sqrt{144 \times 10}}{\sqrt{225 \times 10}}$$

$$= \frac{\sqrt{144}}{\sqrt{225}}$$

$$= \frac{\sqrt{(2 \times 2) \times (2 \times 2) \times (3 \times 3)}}{\sqrt{(3 \times 3) \times (5 \times 5)}}$$

$$= \frac{2 \times 2 \times 3}{3 \times 5}$$

$$= \frac{12}{15}$$

$$= 0.8$$

$$\sqrt{\frac{14.4}{22.5}} = 0.8$$

Question 7(ii)

Evaluate:

$$\sqrt{\frac{0.225}{28.9}}$$

Answer

$$\sqrt{\frac{0.225}{28.9}}$$

$$= \sqrt{\frac{0.225 \times 10000}{28.9 \times 10000}}$$

$$= \frac{\sqrt{225 \times 10}}{\sqrt{289 \times 1000}}$$

$$= \frac{\sqrt{225}}{\sqrt{28900}}$$

$$= \frac{\sqrt{(3 \times 3) \times (5 \times 5)}}{\sqrt{(2 \times 2) \times (5 \times 5) \times (17 \times 17)}}$$

$$= \frac{3 \times 5}{2 \times 5 \times 17}$$

$$= \frac{15}{170}$$

$$= \frac{3}{34}$$

$$\sqrt{\frac{0.225}{28.9}} = \frac{3}{34}$$

Question 7(iii)

Evaluate:

$$\sqrt{\frac{25}{32} \times 2\frac{13}{18} \times 0.25}$$

Answer

$$\sqrt{\frac{25}{32} \times 2\frac{13}{18} \times 0.25}$$

$$= \sqrt{\frac{25}{32} \times \frac{49}{18} \times \frac{25}{100}}$$

$$= \sqrt{\frac{(5 \times 5) \times (7 \times 7) \times (5 \times 5)}{(2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (2 \times 2) \times (5 \times 5)}}$$

$$= \frac{5 \times 7 \times 5}{2 \times 2 \times 2 \times 2 \times 3 \times 5}$$

$$= \frac{175}{240}$$

$$= \frac{35}{48}$$

$$\sqrt{\frac{25}{32} \times 2\frac{13}{18} \times 0.25} = \frac{35}{48}$$

Question 7(iv)

Evaluate:

$$\sqrt{1\frac{4}{5} \times 14\frac{21}{44} \times 2\frac{7}{55}}$$

Answer

$$\sqrt{1\frac{4}{5} \times 14\frac{21}{44} \times 2\frac{7}{55}}$$

$$= \sqrt{\frac{9}{5} \times \frac{637}{44} \times \frac{117}{55}}$$

$$= \sqrt{\frac{(3 \times 3) \times (7 \times 7) \times (13 \times 13) \times (3 \times 3)}{(5 \times 5) \times (2 \times 2) \times (11 \times 11)}}$$

$$= \frac{3 \times 7 \times 13 \times 3}{5 \times 2 \times 11}$$

$$= \frac{819}{110}$$

$$= 7\frac{49}{110}$$

$$\sqrt{1\frac{4}{5} \times 14\frac{21}{44} \times 2\frac{7}{55}} = 7\frac{49}{110}$$

Question 8(i)

Evaluate:

$$\sqrt{3^2 \times 6^3 \times 24}$$

Answer

$$\sqrt{3^2 \times 6^3 \times 24}$$

$$= \sqrt{3 \times 3 \times 6 \times 6 \times 6 \times 24}$$

$$= \sqrt{(3 \times 3) \times (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (2 \times 2) \times (3 \times 3)}$$

$$= 3 \times 2 \times 2 \times 3 \times 2 \times 3$$

$$= 216$$

$$\sqrt{3^2 \times 6^3 \times 24} = 216$$

Question 8(ii)

Evaluate:

$$\sqrt{(0.5)^3 \times 6 \times 3^5}$$

Answer

$$\sqrt{(0.5)^3 \times 6 \times 3^5}$$

$$= \sqrt{\left(\frac{5}{10}\right)^3 \times 6 \times 3^5}$$

$$= \sqrt{\left(\frac{125}{1000}\right) \times 6 \times 243}$$

$$= \sqrt{\left(\frac{125}{500}\right) \times 3 \times 243}$$

$$= \sqrt{\left(\frac{25}{100}\right) \times 729}$$

$$= \left(\frac{5}{10}\right) \times 27$$

$$= \left(\frac{135}{10}\right)$$

$$= 13.5$$

$$\sqrt{(0.5)^3 \times 6 \times 3^5} = 13.5$$

Question 8(iii)

Evaluate:

$$\sqrt{\left[5 + 2\frac{21}{25}\right] \times \frac{0.169}{1.6}}$$

Answer

$$\sqrt{\left[5 + 2\frac{21}{25}\right] \times \frac{0.169}{1.6}}$$

$$= \sqrt{\left[5 + \frac{71}{25}\right] \times \frac{169 \times 10}{16 \times 1000}}$$

$$\begin{aligned}&= \sqrt{\left[\frac{5}{1} + \frac{71}{25}\right] \times \frac{169 \times 10}{16 \times 1000}} \\&= \sqrt{\left[\frac{5 \times 25}{1 \times 25} + \frac{71}{25}\right] \times \frac{169}{16 \times 100}} \\&= \sqrt{\left[\frac{125}{25} + \frac{71}{25}\right] \times \frac{169}{16 \times 100}} \\&= \sqrt{\left[\frac{125 + 71}{25}\right] \times \frac{169}{1600}} \\&= \sqrt{\left[\frac{196}{25}\right] \times \frac{169}{1600}} \\&= \sqrt{\left[\frac{14 \times 14}{5 \times 5}\right] \times \frac{13 \times 13}{40 \times 40}} \\&= \left[\frac{14}{5}\right] \times \frac{13}{40} \\&= \left[\frac{14 \times 13}{5 \times 40}\right] \\&= \left[\frac{182}{200}\right] \\&= \left[\frac{91}{100}\right] \\&= 0.91 \\&\sqrt{\left[5 + 2\frac{21}{25}\right] \times \frac{0.169}{1.6}} = 0.91\end{aligned}$$

Question 8(iv)

Evaluate:

$$\sqrt{5\left[2\frac{3}{4} - \frac{3}{10}\right]}$$

Answer

$$\sqrt{5\left[2\frac{3}{4} - \frac{3}{10}\right]}$$

$$= \sqrt{5\left[\frac{11}{4} - \frac{3}{10}\right]}$$

LCM of 4 and 10 is $2 \times 2 \times 5 = 20$.

$$= \sqrt{5\left[\frac{11 \times 5}{4 \times 5} - \frac{3 \times 2}{10 \times 2}\right]}$$

$$= \sqrt{5\left[\frac{55}{20} - \frac{6}{20}\right]}$$

$$= \sqrt{5\left[\frac{55 - 6}{20}\right]}$$

$$= \sqrt{5\left[\frac{49}{20}\right]}$$

$$= \sqrt{\left[\frac{5 \times 49}{20}\right]}$$

$$= \sqrt{\left[\frac{49}{4}\right]}$$

$$= \sqrt{\left[\frac{7 \times 7}{2 \times 2}\right]}$$

$$= \frac{7}{2}$$

$$= 3\frac{1}{2}$$

$$\sqrt{5\left[2\frac{3}{4} - \frac{3}{10}\right]} = 3\frac{1}{2}$$

Question 8(v)

Evaluate:

$$\sqrt{248 + \sqrt{52 + \sqrt{144}}}$$

Answer

$$\sqrt{248 + \sqrt{52 + \sqrt{144}}}$$

$$= \sqrt{248 + \sqrt{52 + 12}}$$

$$= \sqrt{248 + \sqrt{64}}$$

$$= \sqrt{248 + 8}$$

$$= \sqrt{256}$$

$$= 16$$

$$\sqrt{248 + \sqrt{52 + \sqrt{144}}} = 16$$

Question 9

A man, after a tour, finds that he had spent every day as many rupees as the number of days he had been on tour. How long did his tour last, if he had spent in all ₹1,296?

Answer

Let number of days he had been on tour be x .

Rupees spent every day = number of days he had been on tour = x

Total rupees spent = ₹1,296

Hence,

$$x \times x = ₹1,296$$

$$\Rightarrow x^2 = 1,296$$

$$\Rightarrow x = \sqrt{1,296}$$

$$\Rightarrow x = \sqrt{36 \times 36}$$

$$\Rightarrow x = 36$$

So, the man was on tour for 36 days.

Question 10

Out of 745 students, maximum are to be arranged in the school field for a P.T. display, such that the number of rows is equal to the number of columns. Find the number of rows if 16 students were left out after the arrangement.

Answer

Let the number of columns be x .

The number of rows = the number of columns = x

Students arranged in P.T. display = $745 - 16 = 729$

Hence,

$$x \times x = 729$$

$$\Rightarrow x^2 = 729$$

$$\Rightarrow x = \sqrt{729}$$

$$\Rightarrow x = \sqrt{27 \times 27}$$

$$\Rightarrow x = 27$$

Hence, the number of rows = 27.

Question 11

13 and 31 is a strange pair of numbers such that their square 169 and 961 are also mirror images of each other. Find two more such pairs.

Answer

$$12^2 = 144$$

$$21^2 = 441$$

12 and 21 are mirror images of each other.

And, 144 and 441 are mirror images of each other.

$$102^2 = 10404$$

$$201^2 = 40401$$

102 and 201 are mirror images of each other.

And, 10404 and 40401 are mirror images of each other.

12 and 21; 102 and 201 are 2 such pairs.

Question 12

Find the smallest perfect square divisible by 3, 4, 5 and 6.

Answer

LCM of 3, 4, 5 and 6 is $2 \times 2 \times 3 \times 5 = 60$

As 3 and 5 are not in pair. To make it a perfect square, it must be multiplied with $3 \times 5 = 15$.

So, required perfect square = $60 \times 15 = 900$.

Question 13(i)

If $\sqrt{784} = 28$, find the value of $\sqrt{7.84} + \sqrt{78400}$

Answer

$$\sqrt{7.84} + \sqrt{78400}$$

$$= \sqrt{\frac{784}{100}} + \sqrt{784 \times 100}$$

$$= \frac{\sqrt{784}}{\sqrt{100}} + (\sqrt{784} \times \sqrt{100})$$

$$= \frac{28}{10} + (28 \times 10)$$

$$= 2.8 + 280$$

$$= 282.8$$

$$\sqrt{7.84} + \sqrt{78400} = 282.8.$$

Question 13(ii)

If $\sqrt{784} = 28$, find the value of $\sqrt{0.0784} + \sqrt{0.000784}$

Answer

$$\sqrt{0.0784} + \sqrt{0.000784}$$

$$= \sqrt{\frac{784}{10000}} + \sqrt{\frac{784}{1000000}}$$

$$= \frac{\sqrt{784}}{\sqrt{10000}} + \frac{\sqrt{784}}{\sqrt{1000000}}$$

$$= \frac{28}{100} + \frac{28}{1000}$$

$$= 0.28 + 0.028$$

$$= 0.308$$

$$\sqrt{0.0784} + \sqrt{0.000784} = 0.308$$

Exercise 3(B)

Question 1(i)

If $\sqrt{5} = 2.24$; the value of $\sqrt{20}$ is:

1. 1.12
2. 4.48
3. 2.24×4
4. none of the above

Answer

If $\sqrt{5} = 2.24$, then

$$\sqrt{20}$$

$$= \sqrt{4 \times 5}$$

$$= 2\sqrt{5}$$

$$= 2 \times 2.24$$

$$= 4.48$$

Hence, option 2 is the correct option.

Question 1(ii)

If $\sqrt{27.8} = 5.27$; the value of $\sqrt{2780}$ is:

1. 527

2. 52.7

3. 0.527

4. none of the above

Answer

If $\sqrt{27.8} = 5.27$, then

$$\sqrt{2780}$$

$$= \sqrt{27.8 \times 100}$$

$$= 10\sqrt{27.8}$$

$$= 10 \times 5.27$$

$$= 52.7$$

Hence, option 2 is the correct option.

Question 1(iii)

n is the least natural number that must be added to 23 so that the resulting number is a perfect square, the value of n is:

1. 7

2. 2

3. 5

4. -7

Answer

Since $4^2 < 23 < 5^2$,

The least number that must be added to 23 to make the resulting number a perfect square = $5^2 - 23 = 25 - 23 = 2$.

Hence, option 2 is the correct option.

Question 1(iv)

n is the least natural number that must be subtracted from 23 so that the resulting number is a perfect square, the value of n is:

1. 7

2. 2

3. 5

4. -7

Answer

Since $4^2 < 23 < 5^2$,

The least number that must be subtracted from 23 to make the resulting number a perfect square = $23 - 4^2 = 23 - 16 = 7$.

Hence, option 1 is the correct option.

Question 2(i)

Find the square root of 4761.

Answer

Square root of 4761 by division method:

$$\begin{array}{r}
 \\
 \overline{ 47 } \\
 6 \overline{47 } \\
 \underline{-36} \\
 129 \\
 \underline{-1161} \\
 X
 \end{array}$$

$$\sqrt{4761} = 69$$

Question 2(ii)

Find the square root of 7744.

Answer

Square root of 7744 by division method:

$$\begin{array}{r}
 88 \\
 \hline
 8 \overline{) 7744} \\
 \underline{-64} \\
 1344 \\
 \underline{-1344} \\
 X
 \end{array}$$

$$\sqrt{7744} = 88$$

Question 2(iii)

Find the square root of 15129.

Answer

Square root of 15129 by division method:

$$\begin{array}{r}
 123 \\
 \hline
 1 \overline{) 15129} \\
 \underline{-1} \\
 051 \\
 \underline{-44} \\
 729 \\
 \underline{-729} \\
 X
 \end{array}$$

$$\sqrt{15129} = 123$$

Question 2(iv)

Find the square root of 0.2916.

Answer

Square root of 0.2916 by division method:

$$\begin{array}{r}
 0.54 \\
 \hline
 5 \overline{) 0.2916} \\
 \underline{-25} \\
 416 \\
 \underline{-416} \\
 X
 \end{array}$$

$$\sqrt{0.2916} = 0.54$$

Question 2(v)

Find the square root of 0.001225.

Answer

Square root of 0.001225 by division method:

$$\begin{array}{r}
 0.035 \\
 \hline
 3 \overline{) 0.001225} \\
 \underline{-9} \\
 325 \\
 \underline{-325} \\
 X
 \end{array}$$

$$\sqrt{0.001225} = 0.035$$

Question 2(vi)

Find the square root of 0.023104.

Answer

Square root of 0.023104 by division method:

$$\begin{array}{r}
 \begin{array}{c} 0.152 \\ \hline \end{array} \\
 \begin{array}{r}
 1 \quad \begin{array}{r} \overline{0.023104} \\ -01 \\ \hline 131 \\ -125 \\ \hline 604 \\ -604 \\ \hline X \end{array} \\
 25 \\
 302
 \end{array}
 \end{array}$$

$$\sqrt{0.023104} = 0.152$$

Question 2(vii)

Find the square root of 27.3529.

Answer

Square root of 27.3529 by division method:

$$\begin{array}{r}
 5.23 \\
 \hline
 5 \overline{) 27.3529} \\
 \underline{-25} \\
 102 \overline{) 235} \\
 \underline{-204} \\
 1043 \overline{) 3129} \\
 \underline{-3129} \\
 \text{X}
 \end{array}$$

$$\sqrt{27.3529} = 5.23$$

Question 3(i)

Find the square root of 4.2025.

Answer

Square root of 4.2025 by division method:

$$\begin{array}{r}
 2.05 \\
 \hline
 2 \overline{) 4.2025} \\
 \underline{-4} \\
 40 \overline{) 020} \\
 \underline{-0} \\
 405 \overline{) 2025} \\
 \underline{-2025} \\
 \text{X}
 \end{array}$$

$$\sqrt{4.2025} = 2.05$$

Question 3(ii)

Find the square root of 531.7636.

Answer

Square root of 531.7636 by division method:

$$\begin{array}{r}
 \begin{array}{c} 2 \quad 3 \quad . \quad 0 \quad 6 \\ \hline 2 \quad 5 \quad 31.76 \quad 36 \\ -4 \\ \hline 43 \quad 1 \quad 31 \\ -129 \\ \hline 460 \quad 2 \quad 76 \\ -000 \\ \hline 4606 \quad 2 \quad 76 \quad 36 \\ -276 \quad 36 \\ \hline \quad \quad \quad X \end{array}
 \end{array}$$

$$\sqrt{531.7636} = 23.06$$

Question 3(iii)

Find the square root of 0.007225.

Answer

Square root of 0.007225 by division method:

$$\begin{array}{r}
 0.085 \\
 \hline
 8 \overline{) 0.007225} \\
 \underline{-64} \\
 825 \\
 \underline{-825} \\
 X
 \end{array}$$

$$\sqrt{0.007225} = 0.085$$

Question 4(i)

Find the square root of 245 correct to two places of decimal.

Answer

Square root of 245 correct to two places of decimal by division method:

$$\begin{array}{r}
 15.65 \\
 \hline
 1 \overline{) 245.0000} \\
 \underline{-1} \\
 25 \\
 \underline{-125} \\
 306 \\
 \underline{-1836} \\
 3125 \\
 \underline{-15625} \\
 775
 \end{array}$$

$$\sqrt{245} = 15.65$$

Question 4(ii)

Find the square root of 496 correct to three places of decimal.

Answer

Square root of 496 correct to three places of decimal by division method:

$$\begin{array}{r}
 \begin{array}{ccccccc}
 & 2 & 2 & . & 2 & 7 & 1 \\
 \hline
 2 & \overline{) 496.000000} \\
 & \underline{-4} & & & & & \\
 & 42 & & & & & \\
 & & \underline{096} & & & & \\
 & & & \underline{-84} & & & \\
 & & 442 & & & & \\
 & & & \underline{1200} & & & \\
 & & & & \underline{-884} & & \\
 & & & & & \underline{31600} & \\
 & & & & & & \underline{-31129} \\
 & & & & & & & \underline{47100} \\
 & & & & & & & & \underline{44541} \\
 & & & & & & & & & \underline{2559}
 \end{array}
 \end{array}$$

$$\sqrt{496} = 22.271$$

Question 4(iii)

Find the square root of 82.6 correct to two places of decimal.

Answer

Square root of 82.6 correct to two places of decimal by division method:

$$\begin{array}{r}
 9.08 \\
 \hline
 9 \overline{)82.6000} \\
 \underline{-81} \\
 160 \\
 \underline{-000} \\
 16000 \\
 \underline{-14464} \\
 1536
 \end{array}$$

$$\sqrt{82.6} = 9.08$$

Question 4(iv)

Find the square root of 0.065 correct to three places of decimal.

Answer

Square root of 0.065 correct to three places of decimal by division method:

$$\begin{array}{r}
 0.254 \\
 \hline
 2 \overline{)0.065000} \\
 \underline{-4} \\
 250 \\
 \underline{-225} \\
 2500 \\
 \underline{-2016} \\
 484
 \end{array}$$

$$\sqrt{0.065} = 0.254$$

Question 4(v)

Find the square root of 5.2005 correct to two places of decimal.

Answer

Square root of 5.2005 correct to two places of decimal by division method:

$$\begin{array}{r}
 2.28 \\
 \hline
 2 \overline{) 5.2005} \\
 \underline{-4} \\
 120 \\
 \underline{-84} \\
 3605 \\
 \underline{-3584} \\
 21
 \end{array}$$

$$\sqrt{5.2005} = 2.28$$

Question 4(vi)

Find the square root of 0.602 correct to two places of decimal.

Answer

Square root of 0.602 correct to two places of decimal by division method:

$$\begin{array}{r}
 0.77 \\
 \hline
 7 \overline{) 0.6020} \\
 \underline{-49} \\
 1120 \\
 \underline{-1029} \\
 91
 \end{array}$$

$$\sqrt{0.602} = 0.77$$

Question 5(i)

Find the square root of $3\frac{4}{5}$ correct to two decimal places.

Answer

$$\sqrt{3\frac{4}{5}}$$

$$= \sqrt{\frac{19}{5}}$$

$$= \sqrt{3.8}$$

Square root of 3.8 correct to two places of decimal by division method:

$$\begin{array}{r}
 1.94 \\
 \hline
 1 \overline{) 3.8000} \\
 \underline{-1} \\
 29 \\
 \underline{-261} \\
 384 \\
 \underline{-1536} \\
 364
 \end{array}$$

$$\therefore \sqrt{3.8} = 1.94$$

$$\sqrt{3\frac{4}{5}} = 1.94$$

Question 5(ii)

Find the square root of $6\frac{7}{8}$ correct to two decimal places.

Answer

$$\begin{aligned}
 &\sqrt{6\frac{7}{8}} \\
 &= \sqrt{\frac{55}{8}} \\
 &= \sqrt{6.875}
 \end{aligned}$$

Square root of 6.875 correct to two places of decimal by division method:

$$\begin{array}{r}
 2.62 \\
 \hline
 2 \overline{) 6.8750} \\
 \underline{-4} \\
 46 \overline{) 287} \\
 \underline{-276} \\
 522 \overline{) 1150} \\
 \underline{-1044} \\
 106
 \end{array}$$

$$\therefore \sqrt{6.875} = 2.62$$

$$\sqrt{6\frac{7}{8}} = 2.62$$

Question 6(i)

For each of the following, find the least number that must be subtracted so that the resulting number is a perfect square.

796

Answer

Using division method:

$$\begin{array}{r}
 28 \\
 \hline
 2 \overline{) 796} \\
 \underline{-4} \\
 396 \\
 \underline{-384} \\
 12
 \end{array}$$

Clearly, if 12 is subtracted from 796, the remainder will be a perfect square.

Since, $796 - 12 = 784$

and, $\sqrt{784} = 28$

Hence, 12 should be subtracted from 796.

Question 6(ii)

For each of the following, find the least number that must be subtracted so that the resulting number is a perfect square.

1886

Answer

Using division method:

$$\begin{array}{r}
 43 \\
 \hline
 4 \overline{) 1886} \\
 \underline{-16} \\
 286 \\
 \underline{-249} \\
 37
 \end{array}$$

Clearly, if 37 is subtracted from 1886, the remainder will be a perfect square.

Since, $1886 - 37 = 1849$

and, $\sqrt{1849} = 43$

Hence, 37 should be subtracted from 1886.

Question 6(iii)

For each of the following, find the least number that must be subtracted so that the resulting number is a perfect square.

23497

Answer

Using division method:

$$\begin{array}{r}
 153 \\
 \hline
 1 \overline{) 23497} \\
 \underline{-1} \\
 25 \overline{) 134} \\
 \underline{-125} \\
 303 \overline{) 997} \\
 \underline{-909} \\
 88
 \end{array}$$

Clearly, if 88 is subtracted from 23497, the remainder will be a perfect square.

Since, $23497 - 88 = 23409$

and, $\sqrt{23409} = 153$

Hence, 88 should be subtracted from 23497.

Question 7(i)

For each of the following, find the least number that must be added so that the resulting number is a perfect square.

511

Answer

Using division method:

$$\begin{array}{r}
 22 \\
 \hline
 2 \overline{) 511} \\
 \underline{-4} \\
 111 \\
 \underline{-84} \\
 27
 \end{array}$$

Clearly, 511 is greater than 22^2 .

$\therefore$ On adding the required number to 511, we shall be getting 23^2 i.e, 529.

So, the required number = $529 - 511 = 18$.

Hence, 18 should be added to 511.

Question 7(ii)

For each of the following, find the least number that must be added so that the resulting number is a perfect square.

7172

Answer

Using division method:

$$\begin{array}{r}
 84 \\
 \hline
 8 \overline{) 7172} \\
 \underline{-64} \\
 772 \\
 \underline{-656} \\
 116
 \end{array}$$

Clearly, 7172 is greater than 84^2 .

$\therefore$ On adding the required number to 7172, we shall be getting 85^2 i.e, 7225.

So, the required number = $7225 - 7172 = 53$.

Hence, 53 should be added to 7172.

Question 7(iii)

For each of the following, find the least number that must be added so that the resulting number is a perfect square.

55078

Answer

Using division method:

$$\begin{array}{r}
 \begin{array}{ccc} 2 & 3 & 4 \end{array} \\
 \hline
 2 \quad \begin{array}{r} 55078 \\ -4 \end{array} \\
 \hline
 43 \quad \begin{array}{r} 150 \\ -129 \end{array} \\
 \hline
 464 \quad \begin{array}{r} 2178 \\ -1856 \end{array} \\
 \hline
 \quad \quad 322
 \end{array}$$

Clearly, 55078 is greater than 235^2 .

$\therefore$ On adding the required number to 55078, we shall be getting 235^2 i.e, 55225.

So, the required number = $55225 - 55078 = 147$.

Hence, 147 should be added to 55078.

Question 8

Find the square root of 7 correct to two decimal places; then use it to find the

value of $\sqrt{\frac{4 + \sqrt{7}}{4 - \sqrt{7}}}$ correct to three significant digits.

Answer

Square root of 7 correct to two decimal places using division method:

$$\begin{array}{r}
 2.645 \\
 \hline
 2 \overline{) 7.000000} \\
 \underline{-4} \\
 46 \\
 \underline{-276} \\
 524 \\
 \underline{-2096} \\
 5285 \\
 \underline{-26425} \\
 3975
 \end{array}$$

$$\sqrt{7} = 2.645 \approx 2.65$$

$$\sqrt{\frac{4 + \sqrt{7}}{4 - \sqrt{7}}}$$

$$= \sqrt{\frac{(4 + \sqrt{7})(4 + \sqrt{7})}{(4 - \sqrt{7})(4 + \sqrt{7})}}$$

$$= \sqrt{\frac{(4 + \sqrt{7})^2}{4^2 - (\sqrt{7})^2}}$$

$$= \sqrt{\frac{(4 + \sqrt{7})^2}{16 - 7}}$$

$$= \sqrt{\frac{(4 + \sqrt{7})^2}{9}}$$

$$= \frac{(4 + \sqrt{7})}{3}$$

Putting the value of $\sqrt{7}$, we get

$$= \frac{(4 + 2.65)}{3}$$

$$= \frac{(6.65)}{3}$$

$$= 2.22$$

$$\text{Hence, } \sqrt{7} = 2.65 \text{ and } \sqrt{\frac{4 + \sqrt{7}}{4 - \sqrt{7}}} = 2.22.$$

Question 9

Find the value of $\sqrt{5}$ correct to 2 decimal places; then use it to find the square

root of $\frac{3 - \sqrt{5}}{3 + \sqrt{5}}$ correct to 2 significant digits.

Answer

Square root of 5 correct to two decimal places using division method:

$$\begin{array}{r}
 \begin{array}{c} 2.236 \\ 2 \\ 42 \\ 444 \\ 4446 \end{array} \overline{) \begin{array}{c} 2.236 \\ 5.000000 \\ \underline{-4} \\ 100 \\ \underline{-84} \\ 1600 \\ \underline{-1329} \\ 27100 \\ \underline{-26676} \\ 424 \end{array}}
 \end{array}$$

$$\sqrt{5} = 2.236 \approx 2.24$$

$$\frac{3 - \sqrt{5}}{3 + \sqrt{5}}$$

$$= \sqrt{\frac{(3 - \sqrt{5})(3 - \sqrt{5})}{(3 + \sqrt{5})(3 - \sqrt{5})}}$$

$$= \sqrt{\frac{(3 - \sqrt{5})^2}{3^2 - (\sqrt{5})^2}}$$

$$= \sqrt{\frac{(3 - \sqrt{5})^2}{9 - 5}}$$

$$= \sqrt{\frac{(3 - \sqrt{5})^2}{4}}$$

$$= \frac{(3 - \sqrt{5})}{2}$$

Putting the value of $\sqrt{5}$, we get

$$= \frac{(3 - 2.24)}{2}$$

$$= \frac{(0.76)}{2}$$

$$= 0.38$$

$$\text{Hence, } \sqrt{5} = 2.24 \text{ and } \sqrt{\frac{3 - \sqrt{5}}{3 + \sqrt{5}}} = 0.38.$$

Question 10(i)

Find the square root of $\frac{1764}{2809}$.

Answer

$$\sqrt{\frac{1764}{2809}}$$

$$= \frac{\sqrt{1764}}{\sqrt{2809}}$$

Square root of 1764 using division method:

$$\begin{array}{r}
 42 \\
 \hline
 4 \overline{) 1764} \\
 \underline{-16} \\
 164 \\
 \underline{-164} \\
 X
 \end{array}$$

$$\sqrt{1764} = 42$$

Square root of 2809 using division method:

$$\begin{array}{r}
 53 \\
 \hline
 5 \overline{) 2809} \\
 \underline{-25} \\
 309 \\
 \underline{-309} \\
 X
 \end{array}$$

$$\sqrt{2809} = 53$$

$$\therefore \frac{\sqrt{1764}}{\sqrt{2809}} = \frac{42}{53}$$

$$\sqrt{\frac{1764}{2809}} = \frac{42}{53}$$

Question 10(ii)

Find the square root of $\frac{507}{4107}$.

Answer

$$\sqrt{\frac{507}{4107}}$$

$$= \sqrt{\frac{169}{1369}}$$

$$= \frac{\sqrt{169}}{\sqrt{1369}}$$

$$= \frac{13}{37}$$

Square root of 169 using division method:

$$\begin{array}{r} 13 \\ 1 \overline{) 169} \\ \underline{1} \\ 069 \\ \underline{-69} \\ 0 \end{array}$$

$$\sqrt{169} = 13$$

Square root of 1369 using division method:

$$\begin{array}{r}
 37 \\
 \hline
 3 \overline{)1369} \\
 \underline{-9} \\
 469 \\
 \underline{-469} \\
 0
 \end{array}$$

$$\sqrt{1369} = 37$$

$$\therefore \frac{\sqrt{169}}{\sqrt{1369}} = \frac{13}{37}$$

$$\sqrt{\frac{507}{4107}} = \frac{13}{37}$$

Question 10(iii)

Find the square root of $\sqrt{108 \times 2028}$.

Answer

$$\sqrt{108 \times 2028}$$

$$= \sqrt{108} \times \sqrt{2028}$$

$$= \sqrt{3 \times 36} \times \sqrt{3 \times 676}$$

$$= \sqrt{3} \times \sqrt{36} \times \sqrt{3} \times \sqrt{676}$$

$$= \sqrt{3} \times 6 \times \sqrt{3} \times 26$$

$$= (\sqrt{3} \times \sqrt{3}) \times 6 \times 26$$

$$= 3 \times 6 \times 26$$

$$= 468$$

$$\text{Hence, } \sqrt{108 \times 2028} = 468.$$

Question 10(iv)

Find the square root of $0.01 + \sqrt{0.0064}$.

Answer

$$0.01 + \sqrt{0.0064}$$

$$= 0.01 + \sqrt{\frac{64}{10000}}$$

$$= 0.01 + \frac{\sqrt{64}}{\sqrt{10000}}$$

$$= 0.01 + \frac{8}{100}$$

$$= 0.01 + 0.08$$

$$= 0.09$$

$$\text{Hence, } 0.01 + \sqrt{0.0064} = 0.09$$

Question 11(i)

Find the square root of 7.832 correct to 2 decimal places.

Answer

Square root of 7.832 correct to two places of decimal by division method:

$$\begin{array}{r}
 2.798 \\
 \hline
 2 \overline{) 7.832000} \\
 \underline{-4} \\
 47 \overline{) 383} \\
 \underline{-329} \\
 549 \overline{) 5420} \\
 \underline{-4941} \\
 5588 \overline{) 47900} \\
 \underline{-44704} \\
 3196
 \end{array}$$

$$\sqrt{7.832} = 2.798 \approx 2.80$$

Question 11(ii)

Find the square root of 7.832 correct to 2 significant digits.

Answer

Square root of 7.832 correct to two significant digits by division method:

$$\begin{array}{r}
 2.79 \\
 \hline
 2 \overline{) 7.8320} \\
 \underline{-4} \\
 47 \overline{) 383} \\
 \underline{-329} \\
 549 \overline{) 5420} \\
 \underline{-4941} \\
 479
 \end{array}$$

$$\sqrt{7.832} = 2.79 \approx 2.8$$

Question 12

Find the least number which must be subtracted from 1205 so that the resulting number is a perfect square.

Answer

$$\begin{array}{r}
 34 \\
 \hline
 3 \overline{) 1205} \\
 \underline{-9} \\
 64 \overline{) 305} \\
 \underline{-256} \\
 49
 \end{array}$$

$\therefore$ If 49 is subtracted from 1205, the resulting number will be a perfect square.

Hence, 49 should be subtracted from 1205.

Question 13

Find the least number which must be added to 1205 so that the resulting number is a perfect square.

Answer

$$\begin{array}{r} 3 \quad 4 \\ 3 \overline{) 1205} \\ \underline{-9} \\ 305 \\ \underline{-256} \\ 49 \end{array}$$

Clearly, 1205 is greater than 34^2 .

∴ On adding the required number to 1205, we shall be getting 35^2 i.e, 1225

So, the required number = $1225 - 1205 = 20$.

Hence, 20 should be added to 1205.

Question 14

Find the least number which must be subtracted from 2037 so that the resulting number is a perfect square.

Answer

$$\begin{array}{r}
 4 \quad 5 \\
 \hline
 4 \overline{) 2037} \\
 \underline{-16} \\
 437 \\
 \underline{-425} \\
 12
 \end{array}$$

$\therefore$ If 12 is subtracted from 2037, the resulting number will be a perfect square.

Hence, 12 should be subtracted from 2025.

Question 15

Find the least number which must be added to 5483 so that the resulting number is a perfect square.

Answer

$$\begin{array}{r}
 7 \quad 4 \\
 \hline
 7 \overline{) 5483} \\
 \underline{-49} \\
 583 \\
 \underline{-576} \\
 7
 \end{array}$$

Clearly, 5483 is greater than 74^2 .

$\therefore$ On adding the required number to 5483, we shall be getting 75^2 i.e, 5625

So, the required number = $5625 - 5483 = 142$.

Hence, 142 should be added to 5483.

Exercise 3(C)

Question 1(i)

The value of $18^2 - 17^2$ is :

1. 1
2. -1
3. 35
4. none of the above

Answer

According to the property of square numbers, for any natural number n ,

$$(n + 1)^2 - n^2 = (n + 1) + n$$

$$18^2 - 17^2$$

$$= 18 + 17$$

$$= 35$$

Hence, option 3 is the correct option.

Question 1(ii)

The sum of first four odd natural numbers is :

1. 2^3

2. 2^4

3. 4^3

4. 4^4

Answer

As we know that, the sum of first n odd natural numbers $= n^2$.

$$1 + 3 + 5 + 7$$

$$= 4^2$$

$$= (2^2)^2$$

$$= 2^4$$

Hence, option 2 is the correct option.

Question 1(iii)

24^2 has n at its unit place, the value of n is:

1. 4
2. 16
3. 576
4. 6

Answer

According to the property of square numbers, if the digit at the unit's place of a number is 4 or 6, then its square will always have 6 at its unit's place.

Hence, option 4 is the correct option.

Question 1(iv)

A number ends with 5 zeros, the number of zeros in its square will be :

1. 5
2. 25
3. 10
4. 8

Answer

According to the property of square numbers, if a number ends with n zeroes, its square ends with $2n$ zeroes.

So, if a number ends with 5 zeroes, its square ends with $2 \times 5 = 10$ zeroes.

Hence, option 3 is the correct option.

Question 2

Seeing the value of the digit at unit's place, state which of the following can be square of a number ?

(i) 3051

(ii) 2332

(iii) 5684

(iv) 6908

(v) 50699

Answer

According to the property of square numbers, the ending digit of the square of a number is 0, 1, 4, 5, 6 or 9.

Hence, (i) 3051, (iii) 5684 and (v) 50699 can be square of a number.

Question 3

Squares of which of the following numbers will have 1 (one) at their unit's place ?

(i) 57

(ii) 81

(iii) 139

(iv) 73

(v) 64

Answer

According to the property of square number, if a number has 1 or 9 in its unit's place, then square of this number always has 1 at its unit place.

Hence, (ii) 81 and (iii) 139 will have 1 at their unit's place.

Question 4

Which of the following numbers will not have 1 (one) at their unit's place ?

(i) 32^2

(ii) 57^2

(iii) 69^2

(iv) 321^2

(v) 265^2

Answer

According to the property of square number, if a number has 1 or 9 in its unit's place, then square of this number always has 1 at its unit place.

Hence, (i) 32^2 , (ii) 57^2 and (v) 265^2 will not have 1 at their unit's place.

Question 5

Squares of which of the following numbers will not have 6 at their unit's place ?

(i) 35

(ii) 23

(iii) 64

(iv) 76

(v) 98

Answer

According to the property of square number, if a number has 4 or 6 in its unit's place, then square of this number always has 6 at its unit place.

Hence, (i) 35, (ii) 23 and (v) 98 will not have 6 at their unit's place.

Question 6

Which of the following numbers will have 6 at their unit's place :

(i) 26^2

(ii) 49^2

(iii) 34^2

(iv) 43^2

(v) 244^2

Answer

According to the property of square number, if a number has 4 or 6 in its unit's place, then square of this number always has 6 at its unit place.

Hence, (i) 26^2 , (iii) 34^2 and (v) 244^2 will have 6 at their unit's place.

Question 7

If a number ends with 3 zeroes, how many zeroes will its square have ?

Answer

According to the property of square numbers, if a number ends with n zeroes, its square ends with 2n zeroes.

So, if a number ends with 3 zeroes, its square ends with $2 \times 3 = 6$ zeroes.

Question 8

If the square of a number ends with 10 zeroes, how many zeroes will the number have ?

Answer

According to the property of square numbers, if a number ends with n zeroes, its square ends with 2n zeroes.

So, if the square of a number ends with 10 zeroes, the number will have

$$\frac{10}{2} = 5 \text{ zeroes.}$$

Question 9

Is it possible for the square of a number to end with 5 zeroes ? Give reason.

Answer

According to the property of square numbers, if a number ends with n zeroes, its square ends with $2n$ zeroes.

$2n$ will always be an even number.

Hence, it is not possible for a square of a number to end with 5 zeroes.

Question 10

Give reason to show that none of the numbers given below is a perfect square.

(i) 2162

(ii) 6843

(iii) 9637

(iv) 6598

Answer

According to the property of square numbers, a number having 2, 3, 7 or 8 in its unit's place is never a perfect square.

$\therefore$ (i) 2162, (ii) 6843, (iii) 9637 and (iv) 6598 are not perfect square.

Question 11

State, whether the square of the following numbers is even or odd ?

(i) 23

(ii) 54

(iii) 76

(iv) 75

Answer

(i) 23

Digit at ones place = 3

Square of 3 is 9.

So, digit at ones place of square number = 9

Hence, square of 23 is odd.

(ii) 54

Digit at ones place = 4

Square of 4 is 16.

So, digit at ones place of square number = 6

Hence, square of 54 is even.

(iii) 76

Digit at ones place = 6

Square of 6 is 36.

So, digit at ones place of square number = 6

Hence, square of 76 is even.

(iv) 75

Digit at ones place = 5

Square of 5 is 25.

So, digit at ones place of square number = 5

Hence, square of 54 is odd.

Question 12

Give reason to show that none of the numbers 640, 81000 and 3600000 is a perfect square.

Answer

According to the property of square numbers, if a number ends with n zeroes, its square ends with $2n$ zeroes.

$2n$ will always be an even number. The numbers 640, 81000 and 3600000 have 1, 3 and 5 zeroes respectively.

Hence, the numbers 640, 81000 and 3600000 are not a perfect square.

Question 13(i)

Evaluate:

$$37^2 - 36^2$$

Answer

According to the property of square numbers, for any natural number n ,

$$(n + 1)^2 - n^2 = (n + 1) + n$$

$$37^2 - 36^2$$

$$= 37 + 36$$

$$= 73$$

$$\text{Hence, } 37^2 - 36^2 = 73$$

Question 13(ii)

Evaluate:

$$85^2 - 84^2$$

Answer

According to the property of square numbers, for any natural number n,

$$(n + 1)^2 - n^2 = (n + 1) + n$$

$$85^2 - 84^2$$

$$= 85 + 84$$

$$= 169$$

$$\text{Hence, } 85^2 - 84^2 = 169$$

Question 13(iii)

Evaluate:

$$101^2 - 100^2$$

Answer

According to the property of square numbers, for any natural number n ,

$$(n + 1)^2 - n^2 = (n + 1) + n$$

$$101^2 - 100^2$$

$$= 101 + 100$$

$$= 201$$

$$\text{Hence, } 101^2 - 100^2 = 201$$

Question 14(i)

Without doing the actual addition, find the sum of:

$$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23$$

Answer

As we know that, the sum of first n odd natural numbers $= n^2$.

$$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23$$

$$= 12^2$$

$$= 144$$

$$\text{Hence, } 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 = 144.$$

Question 14(ii)

Without doing the actual addition, find the sum of:

$$1 + 3 + 5 + 7 + 9 + \dots + 39 + 41$$

Answer

As we know that, the sum of first n odd natural numbers $= n^2$.

$$1 + 3 + 5 + 7 + 9 + \dots + 39 + 41$$

$$= 21^2$$

$$= 441$$

Hence, $1 + 3 + 5 + 7 + 9 + \dots + 39 + 41 = 441$.

Question 14(iii)

Without doing the actual addition, find the sum of:

$$1 + 3 + 5 + 7 + 9 + \dots + 51 + 53$$

Answer

As we know that, the sum of first n odd natural numbers $= n^2$.

$$1 + 3 + 5 + 7 + 9 + \dots + 51 + 53$$

$$= 27^2$$

$$= 729$$

Hence, $1 + 3 + 5 + 7 + 9 + \dots + 51 + 53 = 729$.

Question 15

Write three sets of Pythagorean triplets such that each set has numbers less than 30.

Answer

$$3^2 + 4^2 = 5^2$$

$$5^2 + 12^2 = 13^2$$

$$10^2 + 24^2 = 26^2$$

Hence, (3,4,5), (5,12,13) and (10,24,26) are three sets of Pythagorean triplets.

Test Yourself

Question 1(i)

A Pythagorean triplet has one number equal to 6. The Pythagorean triplet are :

1. 5, 6, 7

2. 6, 7, 8

3. 4, 6, 8

4. 6, 8, 10

Answer

$$6^2 + 8^2$$

$$= 100$$

$$= 10^2$$

Hence, option 4 is the correct option.

Question 1(ii)

The number of digits in the square root of 1210000 is :

1. 2

2. 4

3. 3

4. 5

Answer

$$\sqrt{1210000}$$

$$= \sqrt{11 \times 11 \times 10^4}$$

$$= 11 \times 10^2$$

$$= 1100 \text{ Number of digits in } \sqrt{1210000} = 4.$$

Hence, option 2 is the correct option.

Question 1(iii)

The greatest 3-digit perfect square number is :

1. 121

2. 961

3. 100

4. 900

Answer

$$\sqrt{961}$$

$$= 31$$

Hence, option 2 is the correct option.

Question 1(iv)

The area of a square plot is 441 m^2 . Its perimeter is :

1. 84 m^2

2. 84 m

3. 21 m

$$4. 21 \text{ m}^2$$

Answer

Area of square plot = l^2

$$l^2 = 441\text{m}^2$$

$$\Rightarrow l = \sqrt{441\text{m}^2}$$

$$\Rightarrow l = 21\text{m}$$

Perimeter of square plot = $4 \times l$

$$= 4 \times 21\text{m}$$

$$= 84\text{m}$$

Hence, option 2 is the correct option.

Question 1(v)

Statement 1: 3675 is not a perfect square.

Statement 2: After grouping into pairs of equal factors of 3675, if we multiply or divide by the unpaired factor (if any) then the product or the quotient becomes a perfect square.

Which of the following options is correct ?

1. Both the statements are true.
2. Both the statements are false.

3. Statement 1 is true, and statement 2 is false.

4. Statement 1 is false, and statement 2 is true.

Answer

Finding prime factors of 3675, we get :

$$\Rightarrow 3675 = 3 \times (5 \times 5) \times (7 \times 7)$$

Since the prime factor 3 is not in pair.

$\therefore$ 3675 is not a perfect square.

So, statement 1 is true.

$$\Rightarrow 3675 = 3 \times (5 \times 5) \times (7 \times 7)$$

Multiplying by 3 on both sides, we get :

$$\Rightarrow 3675 \times 3 = (3 \times 3) \times (5 \times 5) \times (7 \times 7)$$

$$\Rightarrow 11025 = (3 \times 3) \times (5 \times 5) \times (7 \times 7)$$

Since all prime factor are in pair. Therefore, 3675×3 is a perfect square.

$$\Rightarrow 3675 = 3 \times (5 \times 5) \times (7 \times 7)$$

Dividing by 3 on both sides, we get :

$$\Rightarrow \frac{3675}{3} = (5 \times 5) \times (7 \times 7)$$

$$\Rightarrow 1225 = (5 \times 5) \times (7 \times 7)$$

Since all prime factor are in pair. Therefore, $3675 \div 3$ is a perfect square.

So, statement 2 is true.

Hence, option 1 is the correct option.

Question 1(vi)

Assertion (A) : $\sqrt{\frac{121}{361}} = \sqrt{\frac{11}{19} \times \frac{11}{19}} = \frac{11}{19}$

Reason (R) : The square root of a number n is that number which when multiplied by itself gives n as the product.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

The square root of any number n is that number which when multiplied by itself gives n as the product.

$$\Rightarrow \sqrt{n^2} = \sqrt{n \times n} = n$$

So, reason (R) is true.

Solving,

$$\Rightarrow \sqrt{\frac{121}{361}}$$

$$\Rightarrow \sqrt{\frac{11}{19} \times \frac{11}{19}}$$

$$\Rightarrow \sqrt{\frac{11^2}{19^2}}$$

$$\Rightarrow \frac{\sqrt{11^2}}{\sqrt{19^2}}$$

$$\Rightarrow \frac{11}{19}$$

So, assertion (A) is true and reason (R) clearly explains assertion.

Hence, option 1 is the correct option.

Question 1(vii)

Assertion (A) : 49 is a perfect square, when divided by 3 remainder is 1.

Reason (R) : When each of the perfect square numbers 1, 4, 9, is divided by 3, the remainder is always 1.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Finding prime factors of 49, we get :

$$\Rightarrow 49 = (7 \times 7)$$

Since all prime factors are in pair.

$\therefore$ 49 is a perfect square.

49 divided by 3 leaves quotient 16 and remainder 1.

$$\Rightarrow 49 = 3 \times 16 + 1$$

So, assertion (A) is true.

For each perfect squares,

1 divided by 3 leaves quotient 0 and remainder 1.

$$\Rightarrow 1 = 3 \times 0 + 1$$

4 divided by 3 leaves quotient 1 and remainder 1.

$$\Rightarrow 4 = 3 \times 1 + 1$$

16 divided by 3 leaves quotient 5 and remainder 1.

$$\Rightarrow 16 = 3 \times 5 + 1$$

25 divided by 3 leaves quotient 8 and remainder 1.

$$\Rightarrow 25 = 3 \times 8 + 1$$

36 divided by 3 leaves quotient 12 and remainder 0.

$$\Rightarrow 36 = 3 \times 12 + 0$$

Therefore when each of the perfect square numbers 1, 4, 9, is divided by 3, the remainder is not always 1.

So, reason (R) is false.

Hence, option 3 is the correct option.

Question 1(viii)

Assertion (A) : Natural numbers 5, 12 and 13 are Pythagorean triplets as $12^2 + 5^2 = 13^2$.

Reason (R) : For any natural number n , $(n + 1)^2 - n^2 = (n + 1) + n$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

According to assertion,

$$\Rightarrow 12^2 + 5^2 = 13^2$$

Solving L.H.S. of the above equation, we get :

$$\Rightarrow 12^2 + 5^2$$

$$\Rightarrow 144 + 25$$

$$\Rightarrow 169$$

$$\Rightarrow 13 \times 13$$

$$\Rightarrow 13^2$$

Since, L.H.S. = R.H.S.

So, assertion (A) is true.

According to reason,

$$\Rightarrow (n + 1)^2 - n^2 = (n + 1) + n.$$

Solving L.H.S. of the above equation, we get :

$$\Rightarrow (n + 1)^2 - n^2$$

Using formula; $a^2 - b^2 = (a - b)(a + b)$, we get :

$$= [(n + 1) - n][(n + 1) + n]$$

$$= [n + 1 - n][n + 1 + n]$$

$$= 1 \times [(n + 1) + n]$$

$$= (n + 1) + n$$

Since, L.H.S. = R.H.S.

So, reason (R) is true but it does not explain assertion (A).

Hence, option 2 is the correct option.

Question 1(ix)

Assertion (A) : $1 + 3 + 5 + 7 + \dots + 21 = 10^2$.

Reason (R) : The sum of first n odd natural numbers $= n^2$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

The sum of first n odd natural numbers $= n^2$.

So, reason (R) is true.

According to assertion :

$$1 + 3 + 5 + 7 + \dots + 21 = 10^2$$

Solving the L.H.S. of above equation, we get :

$$\Rightarrow 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21$$

The above is a sum of the first 11 odd natural numbers. So, by formula

$$\Rightarrow 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 = 11^2 = 121.$$

So, assertion (A) is false.

Hence, option 4 is the correct option.

Question 2

Express 21^2 as the sum of two consecutive whole numbers.

Answer

$$(21)^2$$

$$= 21 \times 21$$

$$= 441$$

$$= 220 + 221$$

Hence, the sum of 220 and 221 is equivalent to 21^2 .

Question 3

Find the square root of $10\frac{86}{121}$ by prime factorisation method.

Answer

$$\sqrt{10\frac{86}{121}}$$

$$= \sqrt{\frac{1296}{121}}$$

$$= \frac{\sqrt{1296}}{\sqrt{121}}$$

$$= \frac{\sqrt{(2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3)}}{\sqrt{(11 \times 11)}}$$

$$= \frac{2 \times 2 \times 3 \times 3}{11}$$

$$= \frac{36}{11}$$

$$= 3\frac{3}{11}$$

$$\text{Hence, } \sqrt{10\frac{86}{121}} = 3\frac{3}{11}$$

Question 4

Is 336 a perfect square ? If not, find the smallest number by which 336 be multiplied so that the resulting number is a perfect square.

Answer

Finding prime factors of 336

$$336 = (2 \times 2) \times (2 \times 2) \times 3 \times 7$$

Since the prime factor 3 and 7 are not in pair. So we need one more 3 and one more 7 : $3 \times 7 = 21$.

Hence, 336 is not a perfect square and 21 is the smallest number that should be multiplied with 336 so that the product is a perfect square.

Question 5

Find the least number that must be subtracted from 980 so as to get a perfect square. Also, find the square root of the perfect number obtained.

Answer

$$\begin{array}{r}
 31 \\
 \hline
 3 \overline{) 980} \\
 \underline{-9} \\
 080 \\
 \underline{-61} \\
 19
 \end{array}$$

$\therefore$ If 19 is subtracted from 980, the resulting number will be a perfect square.

Hence, 19 should be subtracted from 980 and $\sqrt{961} = 31$.

Question 6

Find the smallest and the greatest 4-digit numbers, each of which is a perfect square.

Answer

The smallest 4-digit number = 1000

$$\begin{array}{r}
 31 \\
 \hline
 3 \overline{) 1000} \\
 \underline{-9} \\
 100 \\
 \underline{-61} \\
 39
 \end{array}$$

As we can see 1000 is not a perfect square.

Clearly, 1000 is greater than 31^2 .

$\therefore$ On adding the required number to 1000, we shall be getting 32^2 i.e, 1024

So, the required number = $1024 - 1000 = 24$.

$1000 + 24 = 1024$ is a perfect square.

The greatest 4-digit number = 9999

$$\begin{array}{r}
 \\
 \overline{99} \\
 9 \overline{99} \\
 \overline{-81} \\
 189 \overline{1899} \\
 \overline{-1701} \\
 \overline{198}
 \end{array}$$

Clearly, 9999 is greater than 99^2 .

$\therefore$ On subtracting the required number to 9999, we shall be getting 99^2 i.e, 9801.

So, the required number = $9999 - 198 = 9801$.

Hence, 1024 and 9801 are the smallest and greatest 4-digit number.

Question 7

Because $\sqrt{1849} = 43$, find the value of :

(i) $\sqrt{0.1849} + \sqrt{18.49}$

(ii) $\sqrt{184900} - \sqrt{4 \times 18.49}$

Answer

(i)

$$\begin{aligned}& \sqrt{0.1849} + \sqrt{18.49} \\&= \sqrt{\frac{1849}{10000}} + \sqrt{\frac{1849}{100}} \\&= \frac{\sqrt{1849}}{\sqrt{10000}} + \frac{\sqrt{1849}}{\sqrt{100}} \\&= \frac{43}{100} + \frac{43}{10} \\&= 0.43 + 4.3 \\&= 4.73\end{aligned}$$

$$\text{Hence, } \sqrt{0.1849} + \sqrt{18.49} = 4.73$$

(ii)

$$\begin{aligned}& \sqrt{184900} - \sqrt{4 \times 18.49} \\&= \sqrt{1849 \times 100} - \sqrt{4 \times \frac{1849}{100}} \\&= \sqrt{1849} \times \sqrt{100} - \sqrt{4} \times \sqrt{\frac{1849}{100}} \\&= 43 \times 10 - 2 \times \frac{43}{10} \\&= 430 - 2 \times 4.3\end{aligned}$$

$$= 430 - 8.6$$

$$= 421.4$$

$$\text{Hence, } \sqrt{184900} - \sqrt{4 \times 18.49} = 421.4$$

Question 8

The product of two numbers is 256. If one number is four times the other, find the numbers.

Answer

Let the first number be x

The second number is four times the other = $4x$

Hence,

$$x \times 4x = 256$$

$$\Rightarrow 4x^2 = 256$$

$$\Rightarrow x^2 = \frac{256}{4}$$

$$\Rightarrow x^2 = 64$$

$$\Rightarrow x = \sqrt{64}$$

$$\Rightarrow x = 8$$

$$\Rightarrow 4x = 4 \times 8 = 32$$

The numbers are 8 and 32.

Question 9

The area of a square plot is 2116 m^2 . A man takes 5 rounds of the boundary of this plot, find the distance covered by him.

Answer

Area of square plot = l^2

$$l^2 = 2116 \text{ m}^2$$

$$\Rightarrow l = \sqrt{2116 \text{ m}^2}$$

$$\Rightarrow l = 46 \text{ m}$$

Perimeter of square plot = $4 \times l$

$$= 4 \times 46 \text{ m}$$

$$= 184 \text{ m}$$

The man take 5 rounds = $5 \times 184 \text{ m} = 920 \text{ m}$.

Hence, the total distance travelled by the man = 920m.

Question 10

Find the smallest square number which is divisible by 6, 9 and 15.

Answer

LCM of 6, 9 and 15 is $2 \times 3 \times 3 \times 5 = 90$

As 2 and 5 are not in pair. To make it a perfect square, it must be multiplied with
 $2 \times 5 = 10$.

$\therefore$ Required perfect square = $90 \times 10 = 900$.

The smallest square number which is divisible by 6, 9 and 15 is 900.

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